

Breast Cancer Prediction Using Machine Learning

A Project- I submitted to the Mahatma Gandhi Central University in partial fulfilment of the requirements for the award of the degree of

**BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE & ENGINEERING
BY**

Rohit Kumar
(MGCU2022CSIT3023)



DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

MAHATMA GANDHI CENTRAL UNIVERSITY, MOTIHARI

BIHAR- 845401, INDIA

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कंप्यूटर विज्ञान और सूचना प्रौद्योगिकी विभाग
Department of Computer Science and Information
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DECLARATION

This is to certify that the project entitled “**Breast Cancer Prediction**” is being submitted to the Department of Computer Science and Information Technology, Mahatma Gandhi Central University, Motihari, Bihar- 845401, India, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering, is a record of bonafide work carried out by me under the supervision of “**Dr. Vipin Kumar & Ms Smriti Jain**”. The matter embodied in the dissertation has not been submitted in part or full to any University or Institution for the award of any other degree or diploma.

During the preparation of this work, I have not used any AI-based tool to write any part of this dissertation report. I take full responsibility for the submitted content including similarity.

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CERTIFICATE

This is to certify that this project entitled “**Breast Cancer Prediction using Machine Learning**” submitted by **Rohit Kumar**, to the Department of Computer Science & Information Technology, Mahatma Gandhi Central University ,Motihari, Bihar - 845401, India, for the award of the degree of **Bachelor of Technology in Computer Science & Engineering**, is a project work carried out by her under the supervision of **Dr. Vipin Kumar & Ms Smriti Jain**.

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Acknowledgement

“Education is the passport to the future, for tomorrow belongs to those who prepare for it today.”

There is always a sense of gratitude that one express towards others for their help and supervision in achieving goals. This BTech project/dissertation is the result of a challenging journey, upon which many people have contributed and given their support. This formal piece of acknowledgment is an attempt to express the feeling of gratitude toward the people who helped me in successfully completing my Project. A special thanks to our honourable Vice Chancellor Mr. Sanjay Srivastava, my academic Dean Prof. Ranjeet Kumar Choudhary, Head of the department Dr. Vikas Pareek, and all the professors of my department for their support. I would also like to place of appreciation to all the friends whose response and coordination was of utmost importance for the project. Above all, no words can express my feelings to my parents, classmates, and all those persons who supported me during my project. I am also thankful to all the respondents whose cooperation and support has helped me a lot in collecting the necessary information.

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Abstract

Breast cancer is a major health concern and one of the leading causes of death among women worldwide. Early detection is essential for improving treatment outcomes and increasing survival rates. This project aims to develop a machine learning-based system to classify breast tumours as either benign (non-cancerous) or malignant (cancerous) using medical data.

The dataset used contains features derived from breast tissue samples, including measurements like radius, texture, perimeter, area, and smoothness. These features provide important information about the structure and behaviour of tumour cells. The data was analysed through exploratory data analysis (EDA) to understand relationships between variables and detect patterns.

Various machine learning models were applied, such as Logistic Regression, Decision Tree, Support Vector Machine (SVM), K-Nearest Neighbours (KNN), and Gradient Boosting. These models were evaluated based on accuracy, precision, recall, and F1-score to determine their effectiveness in predicting breast cancer.

Among all models, Logistic Regression showed the highest performance, offering reliable and accurate classification. This project demonstrates the potential of machine learning in supporting medical professionals by providing fast and data-driven cancer diagnosis.

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7. INTRODUCTION

7.1 BACKGROUND

One of the most prevalent malignancies in women and a leading cause of death globally is breast cancer. It happens when cells in the breast tissue proliferate out of control, creating lumps or tumours that, if left untreated, could be fatal. Early detection is vital for improving survival rates and reducing the risk of complications. Mammograms, ultrasounds, and biopsies are examples of traditional diagnostic techniques that work well but can be expensive, time-consuming, and occasionally subject to human error.

As technology advances, machine learning offers a faster, data-driven approach to assist in diagnosing breast cancer. By analysing patterns in patient data, machine learning models can determine if a tumour is malignant or benign. with high accuracy. These systems can support doctors in making better and quicker decisions, improving the overall healthcare process. Several machine learning algorithms are used in this research. to build a reliable prediction model for early breast cancer detection.

7.2 OBJECTIVE OF PROJECT

This project's primary objective is to develop a machine learning-driven model that, using patient medical data, can reliably determine whether a breast tumour is benign or malignant. This prediction helps in early identification, which is essential for efficient medical care and reducing the risk of cancer spreading.

The specific objectives of the project include:

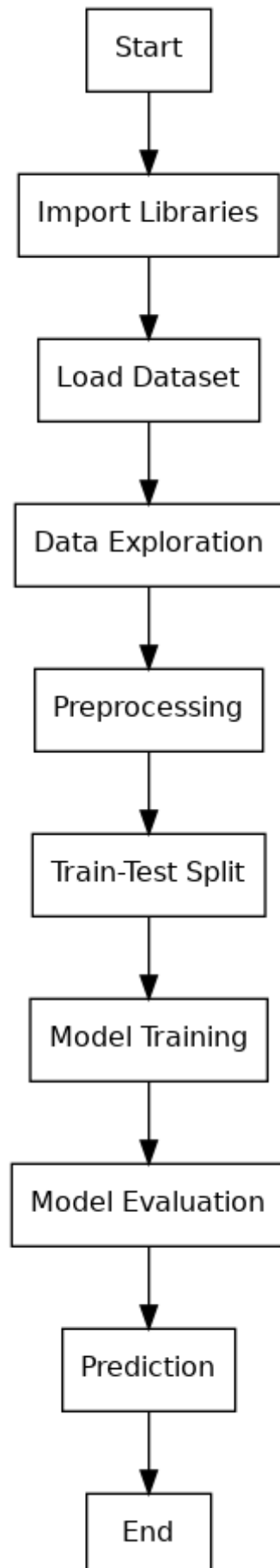
- To study and understand the breast cancer dataset by performing exploratory data analysis (EDA).
- To clean, preprocess, and Get the data ready for machine learning models.
- Implement classification techniques including K-Nearest Neighbours (KNN), Decision Trees, Support Vector Machines (SVM), Logistic Regression, and Gradient Boosting .
- To evaluate these models' respective performances that use the confusion matrix, F1-score, recall, accuracy, and precision. To determine which model performs the best for reliable breast cancer prediction.
- To provide a user-friendly prediction system that could assist healthcare professionals in real-time diagnosis support.

7.3 SCOPE OF THE PROJECT

This breast cancer prediction project has strong potential to create a meaningful impact in the healthcare domain. In the future, several enhancements can be made to improve its performance and usability:

- 1. More Data:** Incorporating larger and more diverse datasets may substantially improve the accuracy of the model and generalizability.
- 2. Instantaneous Prediction:** Developing a web or mobile application will allow real-time predictions, making the tool accessible to healthcare professionals and patients.
- 3. Deep Learning Models:** Implementing advanced techniques like Convolutional Neural Networks (CNNs) can enable image-based diagnosis using mammograms or other scans.
- 4. Model Improvement:** Applying hyperparameter tuning, feature selection, and ensemble methods can further boost model performance.
- 5. Explainable AI:** Integrating explainable AI will help interpret the model's decisions, building trust among medical practitioners.
- 6. Integration with Medical Records:** Linking the system with electronic health records (EHRs) can support early detection and automated screening.

7.4 RESEARCH FLOWCHART



8. INSPIRATION

- One of the main causes of death for women worldwide is breast cancer.
- Despite medical advancements, early detection remains a major challenge.
- Many patients are diagnosed too late due to lack of access to timely screening tools.
- Underdeveloped and rural areas often lack reliable diagnostic facilities.
- Large datasets may be processed and patterns can be found with machine learning efficiently.
- A predictive model can support doctors with fast and accurate diagnosis.
- This project aims to reduce improving early detection and minimizing human error using ML techniques.
- It seeks to create a low-cost, accessible solution for both urban and rural medical systems.
- The ultimate objective is to integrate healthcare and technology to help save lives through smarter decision-making.

9. FUNCTIONALITY

9.1 TECHNOLOGICAL IMPLEMENTATION

This project aims to employ machine learning to forecast breast cancer models. The implementation involved several steps, each utilizing different tools and libraries to prepare, model, and evaluate the data effectively.

1. Data Acquisition and Preparation

- The Pandas was used to import the dataset from a CSV file.
- The information included various features extracted from images of breast masses.
- To facilitate model training, the target variable, diagnosis, was changed from a categorical one (M = Malignant, B = Benign) to a numerical one (0 = Malignant, 1 = Benign).

2. Data Cleaning and Preprocessing

- Irrelevant columns were dropped.
- Missing values, if any, were handled.
- The features and target were separated, The dataset was separated into training and testing sets using the train test split function from Scikit-learn.

3. Data Visualization

- Seaborn and Matplotlib libraries were used to visualize feature correlations, distributions, and class balances.
- Heatmaps and pair plots helped in understanding relationships between variables.

4. Model Building

- Multiple machine learning tests were conducted on algorithms.:

-  Logistic Regression
-  Decision Tree Classifier
-  Nearest Neighbours (KNN)
-  Support Vector Classifier (SVC)
- Gradient Boosting Classifier

- These applied classification strategies were put into practice using Scikit-learn, and instructed with the training dataset..

5. Assessment regarding the Model

- The applied approaches were evaluated use the metrics listed below:
 - Model effectiveness Score
 - Confusion Matrix
 - Classification Report (which includes precision, recall, and F1score)
- These measures shed light on each model's dependability and performance.

6. Development Environment and Tools

- Python was used to implement the project utilizing Google Collab.
- Core libraries used include:
- For data manipulation, use Pandas and NumPy
- For data visualization, use Matplotlib and Seaborn;
- For machine learning modelling assessment, use Scikit learn.

10. PROBLEM STATEMENT

- One of the most common and deadly illnesses impacting women worldwide is breast cancer.
- prompt and precise diagnosis is vital for improving survival rates and enabling effective treatment.
- Conventional diagnostic methods are often manual, time intensive, and subject to human error.
- With the increasing volume of medical data, there is a need for intelligent systems to support faster and more precise diagnosis.
- Machine learning provides advanced techniques to analyse complex datasets, uncover hidden patterns, and make data-driven predictions.
- This project focuses on building a machine learning model that classifies breast tumours as either benign or malignant using clinical features extracted from diagnostic images.
- Features include tumour size, shape, texture, and other measurable attributes derived from medical imaging.
- The goal is to assist healthcare professionals by delivering accurate, efficient, and automated predictions, ultimately reducing diagnostic errors and improving patient care outcomes.

11. DATASET DESCRIPTION

11.1 DATA SOURCE

The dataset utilized for breast cancer detection was obtained from Kaggle, a widely used platform for datasets and data science competitions.

Dataset-Link :- <https://www.kaggle.com>

Features calculated from digital pictures of breast mass fine needle aspiration (FNA) are included in the collection. These qualities, which include radius, roughness, perimeter, area, smoothness, and others, characterize the properties of the cell nuclei that are visible in the picture. The target variable shows if the tumour is malignant or benign.

11.2 DATA PREPROCESSING

An essential part of any machine learning endeavour is preprocessing. It guarantees that the data is consistent, clean, and prepared for algorithmic usage. A thorough yet condensed description of each preparation procedure used on the Breast Cancer Wisconsin dataset can be found below.

1. Removing Unnecessary Columns

- The dataset includes an id column that serves only as a unique identifier for each patient.
- This column does not contribute any useful information for predicting whether a tumour is cancerous or not.
- Action Taken: We removed the id column to avoid including irrelevant data in the model.

2. Converting Categorical Labels to Numbers

- The target column diagnosis contains text values: B for Benign (non-cancerous) , M for Malignant (cancerous).
- Machine learning models work best with numerical data.
- Action Taken: We converted the labels into numbers: $B \rightarrow 0$, $M \rightarrow 1$.
- This made it easier for the model to understand and classify the tumours.

3. Checking for Missing or Null Values

- Missing data can negatively affect the model's performance.

- We checked all columns for missing values or blanks.
- Result: The dataset is complete with no missing values.
- Action Taken: No cleaning was needed in this step.

4. Feature Scaling (Standardization or Normalization)

- The features (like radius mean, area worst, etc.) have different ranges and units.
 - For example, one feature might range from 0– 1, another from 100–1000.
- If features are on different scales, some models may perform poorly or give biased results.
- Action Taken: We applied scaling techniques:
 - Standard Scaler: Adjusts features so they with a standard deviation of 1 and a mean of 0.
 - Min-Max-Scaler: Scales features to a range between 0 and 1.
- This helps improve model accuracy and speed, especially for methods like SVM, KNN, and Logistic Regression.

5. Dataset Splitting

- In order to train and assess our model, we split the dataset into two sections:

- Instructional Set (usually 70–80%): utilized for model training.
- Instructional Set (usually 20–30%): used to assess how well the model performs with unidentified data.

6. Optional Step – Feature Selection

- Not all features contribute equally to prediction.
- Some may be irrelevant or even reduce accuracy if they introduce noise.
- Action Taken (optional but useful): We can apply:
 - Correlation analysis: To identify highly related or duplicate features.
 - Feature importance methods (like Random Forest): To rank features based on their impact on prediction.
- This can help us reduce the dataset size and make the model more efficient.

12. THE MODELS

12.1 Logistic Regression

- **Overview:**

Logistic Regression is a simple and extensively utilized model for classification in binary. It forecasts the likelihood of a class (malignant or benign) using a logistic function (sigmoid).

- **Training:**

The pre-processed dataset was used to train the model. using scaled features. It learned the relationship between the features and diagnosis labels.

- **Prediction:**

Following training, the model used input information to predict if a tumour was benign (1) or malignant (0).

- **Evaluation:**

Accuracy, precision, recall, and the confusion matrix were employed. for evaluation.

- **Accuracy:**

Logistic Regression showed good accuracy, especially because of the data's linear form.

- **Training:**

The data sets used for training and testing were kept apart. On both sets, the model did well.

- **Cross-Validation:**

In order to verify consistency and make sure the model works effectively on various data subsets, K-Fold cross-validation was employed.

12.2 K-Nearest Neighbours (KNN)

- **Overview:**

KNN is a straightforward, non-parametric technique that uses the majority class of its closest neighbours to classify a sample.

- **Training:**

KNN does not train in the traditional sense. It memorizes the training data and uses it during prediction.

- **Prediction:**

When given a new sample, the model checks the 'k' closest data points and assigns the most common class among them.

- **Evaluation:**

Performance was measured using classification metrics and confusion matrix.

- **Accuracy:**

The model gave high accuracy with a well-chosen value of 'k'. Tuning 'k' was important for performance.

- **Testing:**

Testing was done after choosing the best 'k' value. The model showed balanced performance.

- **Cross-Validation:**

Utilizing cross-validation, the most reliable value of 'k' and to prevent overfitting.

12.3 Support Vector Machine (SVM)

- **Overview:**

SVM is a strong classifier that determines the optimal decision boundary (hyperplane) to divide two classes by the greatest amount.

- **Training:**

Utilizing the training set, the model to identify the optimal hyperplane separating benign and malignant tumours.

- **Prediction:**

Once trained, SVM classified fresh data points according to their location on the hyperplane.

- **Evaluation:**

The performance was evaluated using the confusion matrix, recall, accuracy, and precision.

- **Accuracy:**

SVM provided high accuracy, especially with kernel functions that handled non-linear data well.

- **Testing:**

It performed consistently on both training and test sets, with low overfitting.

- **Cross-Validation:**

Used to evaluate SVM performance on different folds and tune parameters like C and kernel type.

12.4 Gradient Boosting

- **Overview:**

- Gradient Boosting is a powerful machine learning model that builds many small decision trees one after another. Each new tree tries to fix the mistakes made by the previous ones. It is useful for making accurate predictions in complex problems like cancer detection.

- **Training:**

During Training, Gradient Boosting model learns from the data by creating small decision trees step by step. Each new tree focuses on correcting the errors made by the previous trees

- **Prediction:**

In prediction, the trained Gradient Boosting model uses all the small trees to make a final decision.

- **Evaluation:**

The output of this model is evaluated by comparing predicted results with actual test data. Accuracy, precision, recall, and F1-score are used to check how well the model has performed.

- **Accuracy:**

Out of all the models, the Gradient Boosting model achieved an accuracy of **94.73%**, which is the second highest.

- **Testing:**
 - In testing, the Gradient Boosting model is applied to new, unseen data to check how well it makes predictions. This helps measure its real-world performance after training.

12.5 Decision Tree

Overview:

The decision tree is a supervised learning technique for applications including regression and classification. It was used in this experiment to categorize tumours as benign (1) or malignant (0). In order to create a tree-like structure with a decision outcome at each leaf node, the model divides the dataset into branches according to feature values.

• Training:

The scaled dataset was employed in the Decision Tree model's training. By applying a criterion such as Gini impurity or entropy (information gain), the algorithm repeatedly chose the

features that offered the best split during training. As a result, the model was able to identify patterns in the data that best distinguished between benign and malignant instances.

- **Prediction:**

After training, the model predicted the class of new tumor samples by traversing the learned decision paths in the tree, moving through nodes based on the input feature values until reaching a terminal leaf that assigned a classification.

- **Evaluation:**

The model's performance was evaluated using standard metrics: accuracy, precision, recall, and the confusion matrix. These helped determine how effectively the model distinguished between the two classes and highlighted any tendency toward overfitting.

- **Accuracy:**

The Decision Tree model achieved high accuracy, especially on the training data. However, it showed signs of potential overfitting, which is a known limitation of basic decision trees when they are not pruned or regularized.

- **Data Splitting:**

Eighty percent of the dataset was utilized for training, while twenty percent was used for testing. The Decision Tree's remarkable performance on the training set was followed by a little decline in accuracy on the testing set, highlighting the necessity of strategies like pruning or tree depth limitation.

- **Cross-Validation:**

K-Fold Cross-Validation was used to guarantee model generalization and dependability. This helped lower the variance associated with a single train-test split and confirmed the consistency of the decision-making process by validating the model's performance across several subsets of the data.

12.6 MODEL SELECTION AND OPTIMIZATION

Building a trustworthy and precise machine learning system for breast cancer prediction requires careful consideration of model selection and optimization. Five supervised learning techniques were used in this experiment to categorize tumours as benign or malignant: Decision Tree , Gradient Boosting, Support Vector Machine (SVM), K-Nearest Neighbours (KNN), and Logistic Regression. The prediction performance and generalization capacity of each model were evaluated.

All models were assessed using common measures including accuracy, precision, recall, F1-score, and the confusion matrix to guarantee an objective comparison. The

Logistic Regression is ideal for this classification challenge since it performed the best overall, with high accuracy and consistent outcomes across various data splits. The performance of each model was validated across several data subsets using cross-validation techniques, particularly K-Fold Cross Validation. This technique verified the consistency of the models' predictions and assisted in lowering overfitting.

Additionally, hyperparameter tuning was carried out to optimize each algorithm:

- Grid Search was used to find the ideal number of neighbours (k) for KNN.
- For optimal performance in SVM, the gamma values, regularization parameter (C), and kernel type were adjusted.

Logistic Regression was chosen as the final model for deployment after a thorough evaluation and optimization process because of its exceptional performance, generalization power, and interpretability.

12.7 Final Model Selection

The best model for predicting breast cancer was determined by implementing and evaluating five supervised machine learning algorithms: Gradient Boosting, Decision Tree, Support Vector Machine (SVM), K- Nearest Neighbours (KNN), and Logistic Regression. Each model was assessed and validated using performance metrics such as accuracy, precision, recall, F1-score, and the confusion matrix. After careful analysis, the Logistic Regression Classifier was found to be the most effective model for this classification task. It continuously attained the best accuracy and showed outstanding generalization to fresh data. The ensemble aspect of the model, which combines several decision trees, enhanced its resilience and capacity to manage intricate patterns in the data.

Hyperparameter tuning, which improved parameters like the number of estimators and tree depth, allowed for even greater increases. K-Fold cross-validation demonstrated consistent performance across several data subsets, confirming the model's dependability.

The best predictive model was determined to be the Logistic Regression model. for the breast cancer classification system created in this study based on these findings.

13. PERFORMANCE METRICS:

Accuracy:

Definition:

The percentage of all cases that were appropriately categorized.

Formula

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

Accuracy of Instances = Total Instances True Negatives plus True Positives

Interpretation: gives a broad indication of how frequently the model is accurate. When one class is more prevalent than the other in an unbalanced dataset, it may be deceptive.

Precision:

Definition: This is the percentage of accurately detected favorable results from all of the model's favourable predictions.

Formula:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Interpretation: It reflects the accuracy of optimistic forecasts. A high number suggests that the model is often accurate in predicting a positive outcome.

Recall (Sensitivity):

Definition: The ratio of actual positive examples that were appropriately categorized as such.

Formula:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Interpretation: evaluates the model's capacity to spot favourable examples. An elevated rate of recall suggests that the majority of positive cases are appropriately represented through the model.

F1-Score:

Definition: The precision and recall harmonic average is represented by the F1-score. It balances the trade-off between the two by merging them into a single statistic. This is how it is computed:

Formula:

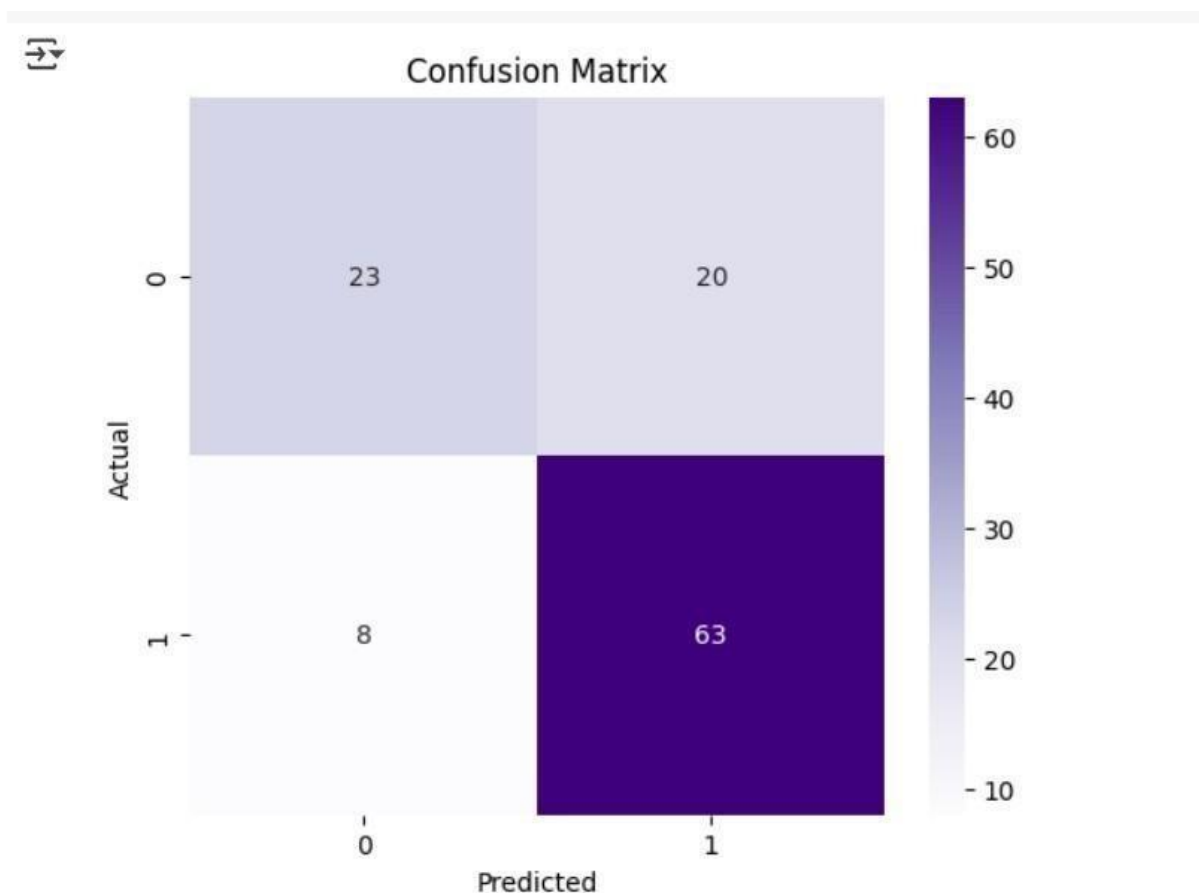
$$\text{F1-Score} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$$

Interpretation: helpful when dealing with unbalanced classes and when you need to achieve equilibrium between precision and recall.

Confusion Matrix:

Definition: A classification-illustrated table the efficacy of the model is illustrated by showing the proportions of false negatives, false positives, true positives, and true negatives.

Interpretation: provides comprehensive details on how the model works and helps to distinguish between the many types of errors that are generated.



14. PERFORMANCE ANALYSIS

Performance study offers a thorough review of the model's ability to forecast wine quality using the dataset. This section highlights strengths and limits, talks about the outcomes of several performance indicators, and makes suggestions for possible areas for improvement.

14.1 INTERPRETATION OF RESULTS

Interpretation of the Performance Metrics

- **Accuracy: 0.956**

The model's overall precision in identifying benign and malignant tumours is known as accuracy. The algorithm accurately predicted the diagnosis in 95 out of 100 cases, with a 95% accuracy rate. The model's great general performance across all classes is reflected in its high accuracy, which makes it a dependable tool for lowering diagnostic errors and aiding in the early diagnosis of breast cancer.

Accuracy
0.956140

- **Precision (0.94)**

Precision quantifies the percentage of cases that are anticipated. as malignant that were actually malignant. With a precision of 94%, the model demonstrates strong reliability in its positive predictions—meaning most of the tumour it identified as cancerous were indeed malignant. This is especially important in medical diagnosis to reduce false positives, where a patient might be incorrectly diagnosed with cancer, leading to unnecessary stress and potentially invasive followup procedures.

Precision
0.945946

- **Recall (0.98)**

Recall calculates the proportion of real cases of malignant tumours that the model was able to detect. The model's 98% recall rate makes it very good at identifying actual cancer cases, which is important for reducing

false negatives, or cases in which a malignant tumour is not diagnosed. A good recall is essential when diagnosing breast cancer since failing to detect a malignant instance can postpone treatment and have a negative impact on patient outcomes.

Recall
0.985915

- **F1 Score (0.965)**

The F1-score combines precision and recall to provide a fair evaluation of a model's accuracy, particularly in problems involving imbalanced categorization. The breast cancer detection model performs well in accurately identifying both malignant and benign instances, as evidenced by its 96% F1-score. Given that early and precise cancer identification can have a substantial impact on patient outcomes, this high score shows that the model successfully reduces both false positives and false negatives.

F1-Score
0.965517

14.2 STRENGTHS AND LIMITATION OF MODEL

Strengths of the Model

The machine learning models developed in this project possess a number of important advantages that enhance their overall efficacy:

1. High Accuracy

Reliability for classification tasks is demonstrated by the models' high accuracy in predicting breast cancer, particularly Logistic Regression and Gradient Boosting.

2. Use of Multiple Algorithms

Several models were used, including Gradient Boosting, SVM, K Nearest Neighbours (KNN), Decision Tree, and Logistic Regression. This made it possible to compare and choose the model with the best performance.

3. Assistance in Making Medical Decisions

The models may help medical practitioners make prompt and precise diagnoses of breast cancer, which is essential for early treatment.

4. Proper Data Preparation

The dataset was carefully cleaned and standardized, which helped improve the performance and reliability of the models.

5. Clear Performance Evaluation

The use of accuracy scores and confusion matrices made it easy to understand how well each model performed, helping to validate the results.

6. Scope for Further Development

The current models can be enhanced in future work, offering a strong foundation for building more advanced diagnostic tools.

14.3 Model Restrictions

Although the machine learning models used in this project showed strong performance, there are some important limitations to consider:

1. Dependence on Clean Data

The models require clean, well-prepared data to work accurately. Missing values, noise, or poorly formatted inputs can negatively affect their predictions.

2. Limited Interpretability

High accuracy is provided by certain models, like Logistic Regression and Gradient Boosting, but do not easily explain how decisions are made. This makes them harder to interpret and less transparent for medical professionals.

3. Narrow Feature Set

The models only use the features available in the dataset. They do not include other useful medical information like patient history, genetic data, or medical images, which could further improve prediction accuracy.

4. No Analysis Over Time

The models handle each case individually and do not account for changes in a patient's condition over time. This limits their usefulness for tracking disease progression.

5. Static Learning Process

After training, the models do not automatically learn from new data. They must be retrained manually, which limits their adaptability in real-world clinical environments.

14.4 Potential Improvements

To enhance the performance and usefulness of the models, the following improvements are recommended:

1. Improved Feature Handling

Applying feature selection techniques or creating new features using domain knowledge can help improve accuracy and reduce overfitting.

2. Hyperparameter Optimization

The ideal model settings for improved performance can be found by using tuning techniques like Grid Search or Randomized Search.

3. Adding Explainability Tools

Tools like SHAP and It is possible to explain individual forecasts using LIME. This is especially important in healthcare, where understanding decisions is critical.

4. Using Ensemble Methods

Combining multiple models (e.g., through boosting or stacking) can improve accuracy and reduce prediction errors.

5. Real-World Integration

Testing the models with real-time clinical data and integrating them into hospital systems would help validate their practical use.

6. Handling Class Imbalance

Applying techniques like SMOTE (Synthetic Minority Oversampling Technique) can help the models perform better when there are fewer malignant cases compared to benign ones.

15. RESULT

The following metrics were derived from the Logistic Regression model's final assessment when looking at the dataset:

- **Performance Accuracy: 95.61%**
- **positive predictive value (precision) of 94.5%**
- **A true positive rate (recall) of 98.5%,**
- **harmonic mean of precision and recall (F1score) of 96.5%.**



Model Comparison Summary:

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.956140	0.945946	0.985915	0.965517
Gradient Boosting	0.947368	0.957746	0.957746	0.957746
Decision Tree	0.921053	0.955882	0.915493	0.935252
KNN	0.754386	0.759036	0.887324	0.818182
SVM	0.622807	0.622807	1.000000	0.767568

Best Model: Logistic Regression (Accuracy: 0.9561)

16. Conclusion

This work aimed to develop a predictive model for classifying breast cancer tumours as either benign or malignant using machine learning techniques. Selecting relevant features, handling missing values, and encoding categorical variables were all part of the dataset's preprocessing. Several classification techniques, such as K-Nearest Neighbours (KNN), Decision Tree, Support Vector Machine (SVM), Gradient Boosting, and Logistic Regression, were used and assessed using F1-score, accuracy, precision, and recall. The comparative analysis revealed that Out of all the models examined, logistic regression performed the most consistently and had the best accuracy. The findings show how well machine learning models enable breast cancer early detection, which is critical for bettering patient care and treatment results.

The promise of data-driven methods in the medical field, particularly for diagnostic applications, is highlighted by this study. To further maximize model accuracy and generalizability, future improvements might incorporate deep learning models, make use of bigger and more varied datasets, and employ more sophisticated feature engineering techniques.

17. REFERENCES

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www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data
2. **YouTube:** Channels like Stat Quest by Josh Starmer offer easy to understand video tutorials on Logistic Regression and other statistical methods.